Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/25/2019
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Capillary Fluidics in Microgravity (Thirsty Walls: A New Paradigm for Air Revitalization in Life Support)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	12/02/2015	**End Date:**	04/09/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 4/9/2020 per NSSC information (Ed., 8/17/21) NOTE: Extended to 12/1/2019 per NSSC information (Ed., 1/18/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AC38G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Large length scale capillary flows in interior corners and complex geometries have an established place in fluids-handling operations in reduced gravity environments. Such flows are driven by wetting conditions and the often complicated geometry of the system. A quantitative and fundamental understanding of corner flows is essential for the myriad fluids management tasks aboard spacecraft including flows in liquid propellant and cryogen storage systems, life support systems, phase-change thermal systems for temperature control and power production, and for materials and biofluids processing.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work aims to provide a silent, low-power method for air quality control aboard long duration spacecraft. For example, a Human mission to Mars needs to have an Environmental Control and Life Support System with unprecedented levels of system reliability and loop closure. This project team asserts that the key to achieving system reliability is to have attributes that favor reliability: ambient temperature, ambient pressure, continuous processes, no reactive chemicals, few rotating components, no safety hazards that need mechanical control. New developments in additive manufacturing and capillary fluid mechanics make it possible to directly expose gases to liquids in microgravity. Terrestrial applications relevant to air revitalization may be cited.
Task Progress:	The manifold 3D printed designs for thin film liquid delivery for smart fluids processes have been completed, designed, fabricated, and demonstrated. A bank of complex test cells were 3D printed and tested demonstrating the over-arching potential of highly controlled bubble-free and rupture-free free surface flows where passive chemical reactions relevant to life support systems aboard spacecraft can take place in a perpetual low-efficiency low-power contactors of functionalized liquids such as ammines (i.e., MEA) and ionic liquids (IL). Our observations add to the confidence in our general approach to both terrestrial and low gravity thin film production via massively manifolded 3D printed conduit networks. We adapted the latest concepts for high surface area fluid delivery to address advanced applications in smart liquids/thirsty walls reactions.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Snyder TJ, Andrews M, Weislogel M, Moeck P, Stone-Sundberg J, Birkes D, Hoffert MP, Lindeman A, Morrill J, Fercak O, Friedman S, Gunderson J, Ha A, McCollister J, Chen Y, Geile J, Wollman A, Attari B, Botnen N, Vuppuluri V, Shim J, Kaminsky W, Adams D, Graf J. "3D systems' technology overview, and new applications in manufacturing, engineering, science, and education." 3D Printing and Additive Manufacturing. 2014 Sep;1(3):169-76. https://doi.org/10.1089/3dp.2014.1502 , Sep-2014
Articles in Peer-reviewed Journals	Weislogel MM, Chen Y, Masica WJ, Kohl FJ, Green RD. "Fifty-Plus-Year postflight analysis of first fluid experiment aboard a spacecraft." AIAA Journal. 2017 Dec;55(12):4042-52. https://doi.org/10.2514/1.J055484 , Dec-2017
Articles in Peer-reviewed Journals	Avhad A, Tan H, Al-Jubaree T, Torres L, Weislogel M. "A numerical investigation of puddle jumping." Physics of Fluids. 2020 Jan;32(1):012109. https://doi.org/10.1063/1.5139235 , Jan-2020
Articles in Peer-reviewed Journals	Weislogel MM, McCraney JT. "The symmetric draining of capillary liquids from containers with interior corners." Journal of Fluid Mechanics. 2019 Jan 25;859: 902-20. https://doi.org/10.1017/jfm.2018.848 , Jan-2019
Articles in Peer-reviewed Journals	Torres LJ, Weislogel MM. "The ejection of large non-oscillating droplets from a hydrophobic wedge in microgravity." npj Microgravity. 2021 Dec 17;7(1):52. https://doi.org/10.1038/s41526-021-00182-4 ; PMID: 34921146 , Dec-2021